## Example 6.1

For the measures in the figure, compute sine, cosine and tangent ratios of the angle $\theta$.

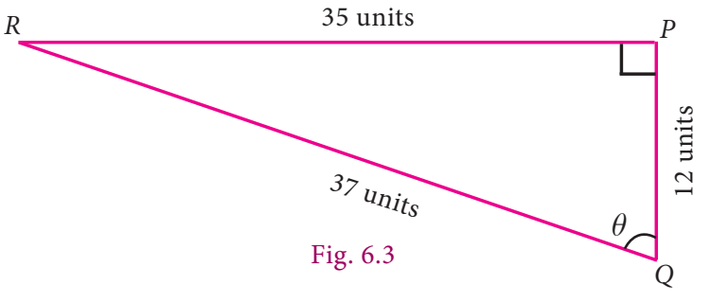


Fig. 6.3

### *Solution*

In the given right angled triangle, note that for the given angle $\theta$, $PR$ is the 'opposite' side and $PQ$ is the 'adjacent' side.

$$\sin\theta = \frac{opposite\,side}{hypotenuse} = \frac{PR}{QR} = \frac{35}{37}$$

$$\cos\theta = \frac{adjacent\,side}{hypotenuse} = \frac{PQ}{QR} = \frac{12}{37}$$

$$\tan\theta = \frac{opposite\,side}{adjacent\,side} = \frac{PR}{PQ} = \frac{35}{12}$$

It is enough to leave the ratios as fractions. In case, if you want to simplify each ratio neatly in a terminating decimal form, you may opt for it, but that is not obligatory.

## Note

- Since trigonometric ratios are defined in terms of ratios of sides, they are unitless numbers.
- Ratios like $\sin v$, $\cos v$, $\tan v$ are not to be treated like $(\sin)\times(v)$, $(\cos)\times(v)$,$(\tan)\times(v)$.

## Thinking Corner

The given triangles $ABC$,$DEF$ and $GHI$ have measures 3-4-5, 6-8-10 and 12-16-20.

Are they all right triangles?

How do you know?

The angles at the vertices $B$, $E$ and $H$ are of equal size (each angle is equal to θ).

With these available details, fill up the following table and comment on the ratios that you get.

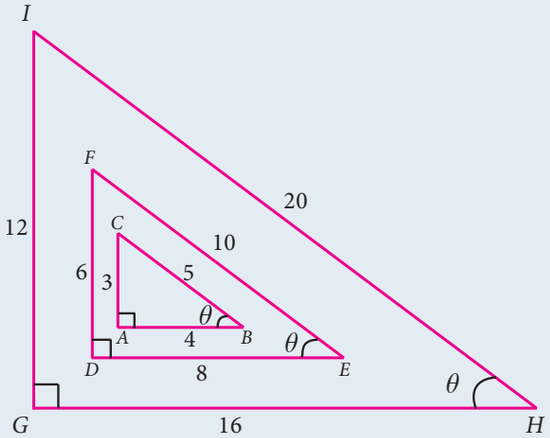


Fig. 6.4



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